Band-1 Hadamard Order 3748 — Miyamoto (Double-seed H_936) + Paley-II; Gap 2052 Still OPEN

H_468/H_936/C_938/H_3748 PASS; 3748 in SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3716_Miyamoto`; `2026-09-15_Band1_Hadamard_3316_Miyamoto` (Miyamoto + Paley-II pattern).
Lab. `/workspace/experiments/band1/hadamard_3748_miyamoto/`.
Cutoff. 2026-09-15T13:05:17Z (~9:05AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 468, 936, and 3748 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double-seed H_{936} = Sylvester-double(H_{468}) where H_{468} is Paley type I ($q = 467$), and independently via Paley type II. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3748 **enters SPK via Paley-II** ($q = 1873$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 3748$): Paley-I **FAIL** ($q = 3747$ not prime); Kronecker **FAIL** ($3748 = 4 \cdot 937$, 937 not Hadamard-admissible). Seed H_{936} uses Sylvester double of Paley-I ($q = 467$); Paley-II seed for 936 **FAIL** ($q = 467 \equiv 3 \pmod 4$). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3716 on box (Miyamoto-only). This increment ships **order 3748** ($3748 = 4 \cdot 937$, 937 prime $\equiv 1 \pmod 4$) via the **Double-seed** pattern plus a classical **Paley-II SPK path** (mirroring 3316, not 3716). Seed H_{468} via Paley type I ($q = 467$); H_{936} = Sylvester-double(H_{468}); symmetric Paley conference C_{938} over prime 937; H_{3748} = Miyamoto(H_{936}, C_{938}). Independently, H_{3748} via Paley type II ($q = 1873$). Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 3748 **is in SPK** via Paley-II (not Miyamoto-only). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3748; Miyamoto; Sylvester doubling; Paley type I; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{468} constructed and verified (Paley type I, $q = 467$). 2. H_{936} constructed and verified (Sylvester double of H_{468}). Paley-II seed for 936 **rejected**. 3. C_{938} constructed and verified (Paley conference, $q = 937$): $CC^\top = 937I$, zero diagonal. 4. H_{3748} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. H_{3748} also constructed and verified (Paley type II, $q = 1873$) — SPK classical path. 6. SPK: $3748 \in$ SPK via Paley-II. Paley-I/Kronecker paths for 3748 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3748 = 4 \cdot 937$
$q = 937$	prime, $937 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{936}
Needed conference	order $938 = q + 1$
H_{468} path	Paley type I, $q = 467$ (prime $\equiv 3 \pmod 4$)
H_{936} path (used)	Sylvester double of H_{468}
H_{936} Paley-II	**REJECTED**: $2(q' + 1) = 936 \Rightarrow q' = 467$ prime but $\equiv 3 \pmod 4$
Conference path	Symmetric Paley over prime 937
Paley-I for 3748	**REJECTED**: $q' = 3747$ not prime
Paley-II for 3748	**PASS**: $q' = 1873$ prime $\equiv 1 \pmod 4$; $2(q' + 1) = 3748$
Kronecker for 3748	**REJECTED**: only 4×937; $937 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{468}	Paley type I, $q = 467$	0	0.1504	**PASS**
H_{936}	Sylvester double of H_{468}	0	0.3591	**PASS**
C_{938}	Paley conference, $q = 937$	0	0.7976	**PASS**
H_{3748}	Miyamoto(H_{936}, C_{938})	0	22.9396	**PASS**
H_{3748}	Paley type II, $q = 1873$	0	25.1465	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh + Paley-II) $\approx$ 51.1921 s. Compressed ± 1 matrices saved as `H468_pm1.npz`, `H936_pm1.npz`, `C938_pm1.npz`, `H3748_pm1.npz`, `H3748_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior ships (through 3716/928/464/...; **exclude** 3716/3524 Miyamoto-only) $\cup$ verified H_{468}, H_{936} $\cup$ Paley-II order 3748 $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3748 enters SPK by the classical Paley-II path.

Quantity	Value
Known count	886
3748 in SPK	**True** (via Paley-II)
936 in SPK	**True** (Sylvester double of Paley-I)
468 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**No** (in SPK via Paley-II; like 3316)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

28/28 PASS (Sylvester/Paley/Miyamoto controls; $q = 467/937/1873$ arithmetic; Double path for H_{936}; Paley-II seed path for 936 rejected; SPK path eligibility for 3748 via Paley-II; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 3748 is **in SPK via Paley-II** (Miyamoto also verified; not Miyamoto-only).

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 977 \to n = 3908$ Double-seedable) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_3748_miyamoto/` · *Script:* `verify_hadamard_3748.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.